# FSM DESIGN

**AIM** - To design a two-state Moore FSM that controls the operation of a 4-bit counter based on the input run. The FSM determines whether the counter increments or retains its current value at each active clock edge.

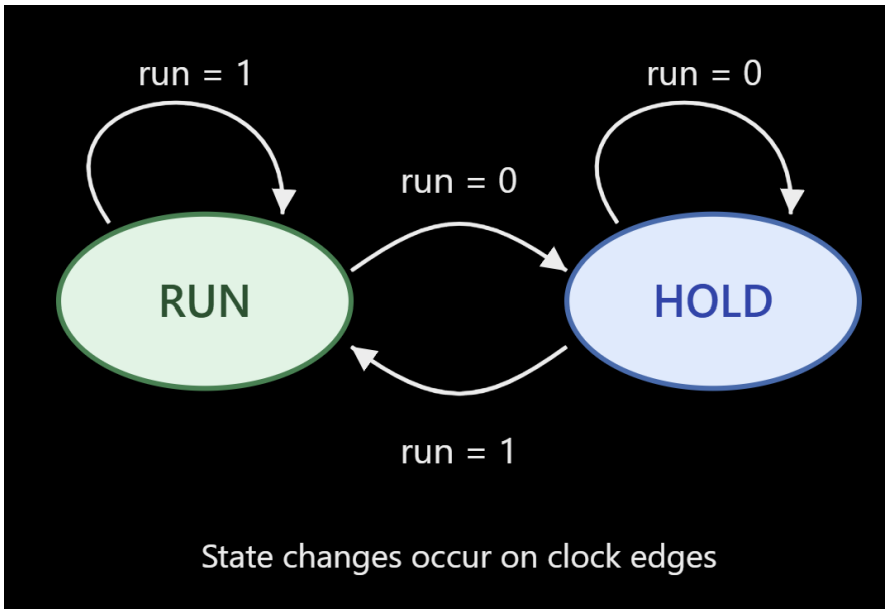
FSM has only two states:

- RUN: The counter is enabled and increments by one at every rising edge of the clock.
- HOLD: The counter is disabled and retains its previous value.

The input run determines the state:

| Run input | FSM state | Counter behaviour |
|---|---|---|
| 1 | RUN | Increment by 1 |
| 0 | HOLD | Keep the current count |

## State diagram




RUN → RUN (run = 1): The FSM stays in RUN while the counter continues incrementing.

RUN → HOLD (run = 0): The FSM moves to HOLD when counting is disabled.

HOLD → HOLD (run = 0): The FSM stays in HOLD and the count remains unchanged.

HOLD → RUN (run = 1): The FSM returns to RUN and counting resumes.

## transition table

The transition table shows the current state, input, and next state for every possible combination.

| Current state | run | Next state | Meaning |
|---|---|---|---|
| HOLD | 0 | HOLD | Remain paused |
| HOLD | 1 | RUN | Resume counting |
| RUN | 0 | HOLD | Pause counting |
| RUN | 1 | RUN | Continue counting |

Assume the following state encoding:

- HOLD = 0
- RUN = 1

Let S be the current state and S+ the next state

| S | run | S+ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

The next-state equation is therefore:

S+ = run

The equation means that at the next active clock edge, the FSM enters RUN if run = 1, or HOLD if run = 0

## Moore FSM output logic

A Moore FSM is a finite state machine whose outputs depend on its current state, rather than directly on its external inputs.

| Current state | State encoding | Counter enable E |
|---|---|---|
| HOLD | 0 | 0 |
| RUN | 1 | 1 |

E = S

- When E=1, the counter is enabled.
- When E=0, the counter holds its previous value.

The input run controls the next state, while the current state controls the Moore output.

## Synchronous reset behaviour

reset = 1: At the next rising clock edge, the FSM is set to HOLD.

reset = 0: The FSM operates normally according to the run input.